

1. What is an Artificial Neural Network (ANN)?

An Artificial Neural Network is a computer model inspired by the human brain. It consists of interconnected neurons (nodes) that process data and learn patterns from examples.

Example:

- Face recognition
 - Speech recognition
 - Spam email detection
-

2. Why are ANNs inspired by the human brain?

Because the human brain learns from experience using neurons connected together. ANN tries to mimic this learning process by connecting artificial neurons.

Brain vs ANN:

- Brain → Biological neurons
 - ANN → Artificial neurons
-

3. What are the main components of a neuron?

An artificial neuron mainly has:

1. **Inputs (x)** → Data given to neuron
2. **Weights (w)** → Importance of each input
3. **Bias (b)** → Extra value to adjust output
4. **Summation Function** → Adds weighted inputs
5. **Activation Function** → Produces final output

Formula:

$$y = f(\sum wx + b)$$

4. What is the role of weights and bias in ANN?

Weights

- Decide importance of inputs
- Higher weight = more influence

Bias

- Helps shift output
- Improves learning flexibility

Simple Example:

Marks prediction:

- Study hours may have higher weight
 - Attendance may have lower weight
-

5. What is an activation function?

Activation function decides whether a neuron should activate or not.

It converts input into output.

Common Activation Functions:

1. Sigmoid
 2. ReLU
 3. Tanh
 4. Softmax
-

6. Why do we need activation functions in neural networks?

Without activation functions:

- Neural network behaves like simple linear model
- Cannot solve complex problems

Activation functions introduce **non-linearity**.

Example:

Image recognition needs non-linear decision making.

7. Differentiate between biological neuron and artificial neuron

Biological Neuron	Artificial Neuron
Present in human brain	Present in computer
Uses electrochemical signals	Uses mathematical calculations
Learns naturally	Learns from data
Very complex	Simplified model

8. What is supervised learning?

In supervised learning:

- Input and output both are given
- Model learns from labeled data

Example:

- Email → Spam or Not Spam

Formula Idea:

Input → Correct Answer → Learn

9. What is unsupervised learning?

In unsupervised learning:

- Only input data is given
- No correct output labels

Model finds hidden patterns automatically.

Example:

Customer grouping in shopping apps.

10. What is reinforcement learning?

In reinforcement learning:

- Agent learns using rewards and penalties

Example:

- Robot learning to walk
- Game playing AI

Correct action → Reward

Wrong action → Penalty

11. What is a single-layer perceptron?

It is the simplest neural network having:

- One input layer
- One output layer
- No hidden layer

Used for:

Simple linear classification problems.

12. What is a multilayer perceptron (MLP)?

MLP is a neural network with:

- Input layer
- One or more hidden layers
- Output layer

It can solve complex non-linear problems.

13. What are input, hidden, and output layers?

Input Layer

Receives data.

Hidden Layer

Processes and learns features.

Output Layer

Gives final prediction/result.

Example:

Image → Hidden processing → Cat/Dog output

14. Why are hidden layers important?

Hidden layers:

- Learn complex patterns
- Extract features
- Improve prediction accuracy

Without hidden layers:

- ANN cannot solve difficult problems.
-

15. What is deep learning?

Deep learning is a part of ANN having many hidden layers.

Applications:

- Self-driving cars

- Chatbots
 - Image recognition
-

16. Differentiate between shallow and deep neural networks

Shallow Network	Deep Network
Few hidden layers	Many hidden layers
Simple learning	Complex learning
Less accuracy	Higher accuracy
Faster training	Slower training

17. What is the purpose of neurons in hidden layers?

Hidden layer neurons:

- Extract important features
- Detect patterns
- Transform input into useful information

Example:

In face detection:

- First layer → edges
 - Next layer → eyes/nose
 - Final layer → full face
-

18. What is a feedforward neural network?

In feedforward network:

- Data moves only in one direction
- Input → Hidden → Output

No feedback connection exists.

19. What is recurrent feedback in neural networks?

Recurrent feedback means:

- Output is fed back into the network again

Used in:

- Sequence data
- Time-series prediction
- Language translation

Example:

RNN (Recurrent Neural Network)

20. What are the limitations of a single-layer perceptron?

Limitations:

1. Can solve only linear problems
2. Cannot solve XOR problem
3. Lower accuracy for complex tasks
4. No hidden layer for deep learning

XOR Problem:

$$XOR(0,0) = 0, XOR(0,1) = 1, XOR(1,0) = 1, XOR(1,1) = 0$$

Single-layer perceptron cannot classify this properly because it is non-linear.

21. Explain the sigmoid activation function

Sigmoid is an activation function that converts input values between **0 and 1**.

Formula:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Features:

- Produces probability-like output
- Used in binary classification

Example:

Spam detection:

- Output near 1 → Spam
 - Output near 0 → Not spam
-

22. What are the advantages and disadvantages of sigmoid?

Advantages:

1. Output between 0 and 1
2. Smooth curve
3. Useful for probability prediction

Disadvantages:

1. Causes vanishing gradient problem
 2. Training becomes slow
 3. Not zero-centered
 4. Computationally expensive
-

23. What is the ReLU activation function?

ReLU means **Rectified Linear Unit**.

It outputs:

- 0 for negative input
- Same value for positive input

Formula:

$$f(x) = \max(0, x)$$

Example:

- Input = -3 → Output = 0
 - Input = 5 → Output = 5
-

24. Why is ReLU widely used?

Reasons:

1. Faster training
2. Simple calculation
3. Reduces vanishing gradient problem
4. Efficient for deep learning

Used in:

- CNN
 - Deep neural networks
 - Image processing
-

25. What is the tanh activation function?

Tanh (Hyperbolic Tangent) converts values between **-1 and +1**.

Formula:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Features:

- Zero-centered output
 - Better than sigmoid in many cases
-

26. Differentiate between sigmoid and tanh

Sigmoid	Tanh
Output range: 0 to 1	Output range: -1 to 1
Not zero-centered	Zero-centered
Slower learning	Faster learning
More vanishing gradient issue Less compared to sigmoid	

27. What is softmax activation?

Softmax converts outputs into probabilities.

Formula:

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Properties:

- Output probabilities sum to 1
 - Used for multi-class classification
-

28. In which applications is softmax used?

Applications:

1. Image classification
2. Digit recognition
3. Language processing
4. Multi-class prediction

Example:

Animal classification:

- Cat = 0.7
- Dog = 0.2
- Horse = 0.1

Highest probability becomes prediction.

29. What is the vanishing gradient problem?

During backpropagation, gradients become extremely small.

As a result:

- Weights stop updating properly
- Learning becomes very slow

Mostly happens in:

- Deep neural networks
 - Sigmoid/tanh activations
-

30. Which activation functions help reduce vanishing gradients?

Activation functions:

1. ReLU
2. Leaky ReLU
3. ELU

These maintain stronger gradients during training.

31. What is training in ANN?

Training is the process where ANN learns patterns from data by adjusting weights and bias.

Steps:

1. Input data
2. Predict output
3. Calculate error
4. Update weights
5. Repeat

32. What is an epoch?

One complete pass of the entire training dataset through the neural network is called an epoch.

Example:

Dataset = 1000 images

After network sees all 1000 once = 1 epoch

33. What is batch size?

Batch size is the number of training samples processed at one time before updating weights.

Example:

Dataset = 1000 samples

Batch size = 100

Then:

- 100 samples processed together
-

34. What is iteration in ANN training?

One iteration means:

- One batch is processed once.

Formula:

$$\text{Iterations} = \frac{\text{Training Samples}}{\text{Batch Size}}$$

Example:

1000 samples / batch size 100 = 10 iterations per epoch

35. What is forward propagation?

Forward propagation is the process where:

- Input data moves from input layer to output layer
- Predictions are generated

Flow:

Input → Hidden Layer → Output

36. What is backpropagation?

Backpropagation is the process of:

- Sending error backward
- Updating weights to reduce error

Purpose:

Improve model accuracy.

37. Why is backpropagation important?

Importance:

1. Helps network learn
2. Reduces prediction error
3. Optimizes weights and bias
4. Improves accuracy

Without backpropagation:

- ANN cannot train effectively.
-

38. Explain gradient descent

Gradient descent is an optimization algorithm used to minimize error/loss.

It updates weights step-by-step toward minimum error.

Update Formula:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

Where:

- η = learning rate
 - L = loss function
-

39. What is learning rate?

Learning rate controls how much weights are updated during training.

Symbol:

η (eta)

Function:

- Small value $\rightarrow$ Slow learning

- Large value → Fast learning
-

40. What happens if the learning rate is too high or too low?

If learning rate is too high:

1. Training becomes unstable
2. Model may overshoot minimum error
3. Accuracy decreases

If learning rate is too low:

1. Training becomes very slow
2. Takes more epochs
3. May get stuck before optimum solution

Ideal learning rate:

Balanced and moderate value gives best performance.

41. What is loss function?

A loss function measures how much error exists between predicted output and actual output.

Purpose:

- Helps evaluate model performance
- Lower loss means better prediction

Example:

Actual = 10

Predicted = 8

Loss = Error between them

42. Differentiate between loss function and cost function

Loss Function

Error for one training sample

Used for single prediction

Smaller scope

Cost Function

Average error for whole dataset

Used for entire training

Broader scope

Simple Idea:

- Loss = individual error
- Cost = average total error

43. What is Mean Squared Error (MSE)?

MSE is a loss function used mainly in regression problems.

It calculates average squared difference between actual and predicted values.

Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value

Features:

- Squaring removes negative signs
 - Large errors get higher penalty
-

44. What is cross-entropy loss?

Cross-entropy loss measures error in classification problems.

Mostly used in:

- Binary classification
- Multi-class classification

Formula:

$$L = -\sum y \log(\hat{y})$$

Example:

Used in:

- Image classification
 - Spam detection
-

45. What is optimization in ANN?

Optimization means improving the neural network by reducing loss/error.

Goal:

Find best weights and bias values.

Done using:

- Gradient descent
- Adam optimizer
- RMSProp

46. Name different optimization algorithms

Common optimization algorithms:

1. Gradient Descent
2. Stochastic Gradient Descent (SGD)
3. Mini-batch Gradient Descent
4. Adam
5. RMSProp
6. Adagrad

47. What is stochastic gradient descent?

Stochastic Gradient Descent (SGD) updates weights using one training sample at a time.

Features:

- Faster updates
- Less memory usage
- Noisy but efficient learning

48. Differentiate between batch gradient descent and stochastic gradient descent

Batch Gradient Descent	Stochastic Gradient Descent
Uses entire dataset	Uses one sample at a time
Slower	Faster
Stable updates	Noisy updates
More memory required	Less memory required

49. What is overfitting?

Overfitting occurs when model learns training data too perfectly, including noise.

Result:

- High training accuracy
- Poor testing accuracy

Example:

Student memorizes answers but cannot solve new questions.

50. What is underfitting?

Underfitting occurs when model cannot learn patterns properly.

Result:

- Poor training accuracy
- Poor testing accuracy

Causes:

- Too simple model
 - Insufficient training
-

51. Why is data preprocessing important?

Importance:

1. Improves data quality
2. Removes noise
3. Handles missing values
4. Improves model accuracy
5. Speeds up training

Example:

Cleaning duplicate or null values before training.

52. What is normalization?

Normalization scales data into a fixed range, usually 0 to 1.

Formula:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Purpose:

Prevents large values from dominating smaller values.

53. What is feature scaling?

Feature scaling means adjusting feature values to similar ranges.

Methods:

1. Normalization
2. Standardization

Example:

- Age = 25
- Salary = 50000

Without scaling, salary dominates learning.

54. Why do we split data into training and testing datasets?

Training Data:

Used to train the model.

Testing Data:

Used to evaluate performance on unseen data.

Purpose:

Check whether model generalizes properly.

55. What is validation data?

Validation data is used during training to:

- Tune hyperparameters
- Prevent overfitting
- Monitor model performance

Dataset Split:

- Training
 - Validation
 - Testing
-

56. What is one-hot encoding?

One-hot encoding converts categorical data into binary vectors.

Example:**Color Encoded**

Red [1 0 0]

Blue [0 1 0]

Green [0 0 1]

Used because ANN works better with numerical data.

57. What type of data is suitable for ANN?

ANN works best with:

1. Numerical data
2. Large datasets
3. Structured and unstructured data

Examples:

- Images
 - Audio
 - Text
 - Sensor data
-

58. What are categorical and numerical features?**Numerical Features:**

Contain numbers.

Examples:

- Age
- Salary
- Temperature

Categorical Features:

Contain categories/labels.

Examples:

- Gender
- Color

- City
-

59. Why is missing data problematic?

Problems:

1. Reduces accuracy
2. Causes incorrect learning
3. Produces biased predictions
4. Training may fail

Solution:

- Remove missing values
 - Replace using mean/median/mode
-

60. How can ANN handle noisy data?

Methods:

1. Data preprocessing
2. Normalization
3. Regularization
4. Dropout technique
5. Large training data
6. Noise filtering

These improve model robustness and accuracy

61. Which libraries are commonly used for ANN implementation in Python?

Common Python Libraries:

1. NumPy → Numerical operations
 2. Pandas → Data handling
 3. Matplotlib → Data visualization
 4. TensorFlow → Deep learning framework
 5. Keras → High-level ANN API
 6. PyTorch → Deep learning framework
 7. Scikit-learn → Machine learning utilities
-

62. What is TensorFlow?

TensorFlow is an open-source deep learning framework developed by Google.

Uses:

- Building neural networks
- Deep learning
- Image processing
- NLP applications

Features:

- Fast computation
 - GPU support
 - Scalable
-

63. What is Keras?

Keras is a high-level API used for building neural networks easily.

Features:

1. Simple syntax
2. User-friendly
3. Works on TensorFlow backend

Purpose:

Makes ANN implementation easier.

64. Differentiate between TensorFlow and Keras

TensorFlow	Keras
Deep learning framework	High-level API
More complex	Easier to use
Flexible and powerful	Beginner friendly
Backend engine	Runs on TensorFlow

Simple Idea:

- TensorFlow = Engine
- Keras = Easy interface

65. What is PyTorch?

PyTorch is an open-source deep learning library developed by Meta.

Features:

1. Dynamic computation graph
2. Easy debugging
3. Popular in research

Used for:

- Deep learning
 - Computer vision
 - NLP
-

66. How do you create a neural network model in Keras?

Steps:

1. Import libraries
2. Create Sequential model
3. Add Dense layers
4. Compile model
5. Fit model

Example Code:

```
from keras.models import Sequential
from keras.layers import Dense

model = Sequential()

model.add(Dense(10, activation='relu', input_dim=5))
model.add(Dense(1, activation='sigmoid'))
```

67. What is the purpose of model compilation?

Compilation prepares the neural network for training.

During compilation:

- Loss function is selected
- Optimizer is selected

- Performance metrics are defined
-

68. What parameters are specified during model compilation?

Main Parameters:

1. Optimizer
2. Loss function
3. Metrics

Example:

```
model.compile(optimizer='adam',  
              loss='binary_crossentropy',  
              metrics=['accuracy'])
```

69. What is model fitting?

Model fitting means training the model using training data.

Example:

```
model.fit(X_train, y_train, epochs=10)
```

During fitting:

- Forward propagation occurs
 - Error is calculated
 - Backpropagation updates weights
-

70. Explain the use of Sequential() model in Keras

Sequential() is used to create neural networks layer-by-layer in sequence.

Flow:

Input → Hidden → Output

Features:

1. Simple architecture
 2. Easy to build ANN
 3. Best for feedforward networks
-

71. What is the role of Dense layer?

Dense layer means every neuron is connected to every neuron of next layer.

Role:

1. Feature learning
2. Data transformation
3. Pattern recognition

Example:

```
Dense(64, activation='relu')
```

Where:

- 64 = number of neurons
-

72. How do you evaluate a trained model?

Using evaluate() function.

Example:

```
model.evaluate(X_test, y_test)
```

It checks:

- Loss
 - Accuracy
 - Performance on test data
-

73. What is the use of predict() function?

predict() is used to generate predictions for new input data.

Example:

```
model.predict(X_new)
```

Uses:

- Classification
 - Regression
 - Future prediction
-

74. What is accuracy in ANN?

Accuracy measures how many predictions are correct.

Formula:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$

Example:

90 correct out of 100 = 90% accuracy

75. How can you improve model accuracy?

Methods:

1. Increase training data
 2. Data preprocessing
 3. Use better activation functions
 4. Tune hyperparameters
 5. Increase epochs carefully
 6. Prevent overfitting
 7. Use proper optimizer
-

76. Why is differentiation used in backpropagation?

Differentiation calculates gradients/slopes of error.

Purpose:

- Helps determine how weights should change
- Reduces loss function

Without differentiation:

- ANN cannot learn efficiently.
-

77. What is chain rule in ANN?

Chain rule is a differentiation rule used in backpropagation to compute gradients layer-by-layer.

Formula:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Importance:

Helps propagate error backward through multiple layers.

78. Explain weight update rule

Weights are updated to minimize error.

Formula:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

Where:

- η = learning rate
- L = loss function

Purpose:

Improve prediction accuracy.

79. What is the role of derivatives in ANN learning?

Derivatives help:

1. Measure error change
2. Calculate gradients
3. Update weights
4. Optimize learning

They are essential for backpropagation.

80. Why should activation functions be differentiable?

Activation functions should be differentiable because backpropagation requires derivatives to calculate gradients.

Importance:

1. Smooth learning
2. Proper gradient calculation
3. Efficient weight updates

Without differentiability:

- Gradient descent cannot work properly.

81. Explain the ANN experiment you performed.

In the ANN practical, we created and trained a neural network model using Keras and TensorFlow.

Steps performed:

1. Imported dataset
2. Preprocessed data

3. Split data into training and testing sets
4. Created ANN model using Sequential()
5. Added Dense hidden layers
6. Compiled model using optimizer and loss function
7. Trained model using fit()
8. Evaluated accuracy using test data
9. Predicted outputs

Goal:

Train ANN to learn patterns and make predictions.

82. What dataset did you use in your practical?

(Answer according to your practical.)

Common Example Answer:

We used a CSV dataset containing input features and output labels for classification/regression.

Example Datasets:

- Iris dataset
 - MNIST dataset
 - Customer churn dataset
 - Diabetes dataset
-

83. What preprocessing steps were applied?

Preprocessing Steps:

1. Removed missing values
2. Converted categorical data into numerical form
3. Applied normalization/feature scaling
4. Split data into training and testing sets

Purpose:

Improve model performance and training efficiency.

84. Which activation function did you use and why?

Common Answer:

We used **ReLU** in hidden layers because:

1. Faster training
2. Reduces vanishing gradient problem
3. Works well in deep learning

For output layer:

- Sigmoid → Binary classification
 - Softmax → Multi-class classification
-

85. How many hidden layers did your model contain?

Example Answer:

The model contained:

- 1 or 2 hidden layers

Example Architecture:

- Input layer
- Hidden layer 1
- Hidden layer 2
- Output layer

(Modify according to your practical.)

86. Which optimizer did you use?

Common Answer:

We used **Adam optimizer**.

Reason:

1. Fast convergence
2. Adaptive learning rate
3. Better performance

Other optimizers:

- SGD
 - RMSProp
 - Adagrad
-

87. What learning rate was selected?

Example Answer:

A small learning rate such as:

$$\eta = 0.001$$

was selected for stable learning.

Reason:

- Prevents unstable training
 - Gives smooth convergence
-

88. How did you measure performance?

Performance Metrics:

1. Accuracy
2. Loss function
3. Confusion matrix (sometimes)

Example:

```
model.evaluate(X_test, y_test)
```

89. What accuracy did you achieve?

(Answer according to your output.)

Example Answer:

The model achieved around:

- 90% to 98% accuracy

Example:

Accuracy = 95%

90. What challenges did you face during implementation?

Challenges:

1. Data preprocessing issues
2. Overfitting
3. Choosing activation functions
4. Hyperparameter tuning

5. Long training time
 6. Selecting correct epochs and batch size
-

91. Mention real-life applications of ANN.

Applications:

1. Image recognition
 2. Speech recognition
 3. Medical diagnosis
 4. Recommendation systems
 5. Weather forecasting
 6. Autonomous vehicles
 7. Fraud detection
 8. Chatbots
-

92. How is ANN used in image recognition?

ANN learns image features such as:

- Edges
- Shapes
- Patterns

Applications:

- Face recognition
- Object detection
- Handwriting recognition

Mostly CNN (Convolutional Neural Networks) are used.

93. How is ANN useful in medical diagnosis?

ANN helps doctors by:

1. Detecting diseases
2. Analyzing medical images
3. Predicting patient risk
4. Supporting diagnosis

Examples:

- Cancer detection
 - MRI analysis
 - Heart disease prediction
-

94. What is the role of ANN in speech recognition?

ANN converts speech into text by learning voice patterns.

Applications:

- Voice assistants
- Speech-to-text systems
- Virtual assistants

Examples:

- Siri
 - Google Assistant
-

95. How is ANN used in recommendation systems?

ANN analyzes user behavior and preferences to recommend items.

Examples:

- Movie recommendations
- Product suggestions
- Music recommendations

Platforms:

- Netflix
 - Amazon
 - YouTube
-

96. Can ANN be used for weather prediction?

Yes. ANN can analyze historical weather data and predict:

- Rainfall
- Temperature
- Storms

- Climate changes

ANN detects complex weather patterns effectively.

97. How is ANN applied in autonomous vehicles?

ANN helps self-driving cars in:

1. Object detection
2. Lane detection
3. Traffic sign recognition
4. Decision making

Sensors Used:

- Cameras
 - Radar
 - LiDAR
-

98. What are the advantages of ANN?

Advantages:

1. Learns complex patterns
 2. High accuracy
 3. Handles noisy data
 4. Supports parallel processing
 5. Self-learning capability
 6. Works with large datasets
-

99. What are the disadvantages of ANN?

Disadvantages:

1. Requires large data
2. Long training time
3. High computational cost
4. Difficult to interpret
5. Overfitting problem
6. Requires tuning

100. Compare ANN with traditional programming approaches.

ANN	Traditional Programming
Learns from data	Uses fixed rules
Adaptive	Less flexible
Good for complex tasks	Good for simple logical tasks
Requires training	No training required
Handles pattern recognition	Difficult for pattern recognition

Simple Idea:

- Traditional programming → Rules are manually written
- ANN → Learns rules automatically from data